THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Rudolf Virchow	1855
2	**(b)** 0.2 mm	1 mm = 1000 µm
3	**(d)** shrink as it loses water	water leaves by osmosis
4	**(c)** cellulose	carbohydrate
5	**(c)** genes	units of heredity
6	**(b)** apical meristem	tips
7	**(d)** age	one ring per year
8	**(c)** transpiration	creates pull in xylem
9	**(c)** collenchyma	flexible support
10	**(c)** phloem	translocation

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cellulose (1); roughage that aids digestion (1).	1+1
12	A cell without a well-defined nucleus and membrane-bound organelles (1); bacteria (1).	1+1
13	Parenchyma: loosely packed living cells that store food; collenchyma: living cells with unevenly thickened corners that give flexibility (1+1).	1+1
14	Its dead cells have thick, lignified walls (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Chromoplasts give colour to attract pollinators (1.5); leucoplasts store food such as starch (1.5).	1½+1½
16	The epidermis is replaced by cork from cork cambium (1.5); cork protects and reduces water loss (1.5).	1½+1½
17	Large sac filled with cell sap (1); stores water, minerals and wastes (1); maintains turgor pressure (1).	1+1+1
	OR: The vacuoles lose water, turgor pressure falls and the cells become limp (3).	3
18	Squamous (alveoli), cuboidal (kidney tubules), columnar (intestine), ciliated (trachea) (3).	3
	OR: Thin flat cells in alveoli for gas exchange (1.5); columnar with microvilli in the intestine for absorption (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Chloroplasts, chromoplasts, leucoplasts with functions (2). (ii) Own DNA and ribosomes (1). (iii) No; roots have leucoplasts for storage (2).	2+1+2
	OR: (i) They contain actively dividing cells (1). (ii) Cell division (mitosis) (1). (iii) Mitosis and meiosis (1). (iv) Growth, repair, replacement and asexual reproduction (2).	1+1+1+2
20	(i) Squamous, cuboidal, columnar, ciliated, glandular (2). (ii) Squamous in alveoli, columnar in the intestine (2). (iii) Protection, secretion, absorption, sensation (1).	2+2+1
	OR: (i) Blood, bone, cartilage, adipose, areolar, tendons and ligaments (2). (ii) Muscle–bone vs bone–bone (2). (iii) Cells in a fluid matrix (plasma) connecting parts of the body (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 120 days	no nucleus, so no repair
2	**(c)** lysosomes	breaks down worn-out parts
3	**(a)** leucoplasts	storage
4	**(a)** meiosis	four cells, halved chromosomes
5	**(d)** programmed cell death	PCD
6	**(d)** squamous epithelium	gas exchange
7	**(d)** ligaments	elastic tissue at joints
8	**(d)** voluntary skeletal muscles	striated
9	**(a)** hinge joint	movement in one plane
10	**(d)** act as cushions and give flexibility	backbone

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It contains DNA with genes controlling cell activities (2).	2
12	More space for haemoglobin to carry oxygen (1); they cannot divide or repair, so they live only 120 days (1).	1+1
13	Xylem carries water and minerals; phloem carries food (1+1).	1+1
14	Protection, secretion, absorption and sensory functions (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	As size increases, the surface area to volume ratio falls, limiting exchange of materials (3).	3
16	Plasma, red cells, white cells, platelets (1.5); transport of oxygen, nutrients, wastes and defence (1.5).	1½+1½
17	Chromatin condenses into chromosomes (1); chromosomes line up and separate (1); two nuclei form and the cell divides into two identical cells (1).	1+1+1
	OR: Mitosis: one division, two identical cells, growth and repair; meiosis: two divisions, four cells with half the chromosomes, forms gametes (1+1+1).	1+1+1
18	Bone: hard matrix with calcium compounds, strength; cartilage: soft, jelly-like matrix, flexible cushion (1.5+1.5).	1½+1½
	OR: Voluntary: skeletal, e.g. leg muscles; involuntary: smooth and cardiac, e.g. intestine and heart (1.5+1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Nucleus, organelles, size, DNA form (circular vs linear) (3). (ii) Bacteria; plant or animal cell (1). (iii) Nucleoid (1).	3+1+1
	OR: (i) Orderly sequence of growth and division of a cell (1). (ii) Number of divisions, daughter cells, chromosome number, purpose (3). (iii) Growth, repair, reproduction (1).	1+3+1
20	(i) Skeletal, cardiac, smooth (1.5). (ii) Biceps (½). (iii) Cardiac (1). (iv) Movement of arm; pumping blood; moving food (2).	1½+½+1+2
	OR: (i) Features of each: skeletal striated and unbranched; smooth spindle-shaped; cardiac branched, rhythmic (3). (ii) Under conscious control vs not (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** cell	cell theory
2	**(c)** 0.2 mm (200 µm)	5 mm ÷ 25
3	**(b)** increases in weight	water enters by osmosis
4	**(d)** animal cells	animals have no rigid wall
5	**(b)** a well-defined nucleus	nucleoid region
6	**(b)** lateral meristem	cambium
7	**(b)** differentiation	permanent tissue
8	**(d)** increase the surface area for absorption	extensions of epidermal cells
9	**(c)** sclerenchyma	hard and strong
10	**(d)** xylem	tubular, thick-walled

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They release energy from food during respiration and store it as ATP (2).	2
12	They break down worn-out organelles and foreign material using enzymes (2).	2
13	Skeletal, smooth and cardiac (2).	2
14	Tendon connects muscle to bone; ligament connects bone to bone (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Growing cells outside the body in a nutrient medium (1); studying cell function and producing biochemicals/food (2).	1+2
16	They contract rhythmically without fatigue (1.5); they are branched and involuntary, while skeletal muscles are unbranched and voluntary (1.5).	1½+1½
17	Genetically regulated death of selected cells (1.5); essential for development, quality control and immune function (1.5).	1½+1½
	OR: Cells that divide uncontrollably (1); they form tumours and may spread, unlike normal cells that grow, age and die in a controlled manner (2).	1+2
18	Receives stimuli and transmits impulses (2); neuron (1).	2+1
	OR: Ball and socket: shoulder; hinge: knee; pivot: neck; fixed: skull (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Growing cells outside the body (1). (ii) To prevent contamination by microbes (1). (iii) Production of biochemicals or study of cells (1). (iv) The ability of a plant cell to grow into a whole plant (2).	1+1+1+2
	OR: (i) Genetically regulated death of cells (1). (ii) Cancer cells divide without control and spread (2). (iii) To avoid excess cells, tumours and loss of function (2).	1+2+2
20	(i) Where two bones meet (1). (ii) Ball and socket (shoulder), hinge (knee), pivot (neck), fixed (skull) (3). (iii) Ligaments hold bones; tendons attach muscles to bones (1).	1+3+1
	OR: (i) The vertebral column (1). (ii) Cartilage (1). (iii) It expands and contracts (1). (iv) Flexibility, posture, less stress, stronger muscles (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET D

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** mitochondrion	ATP production
2	**(a)** ribosomes	attached to the RER or free
3	**(a)** DNA and ribosomes	evolutionary history
4	**(d)** increase in the number of cells	cell division
5	**(c)** cancer cells	lack of control
6	**(c)** the outer covering and lining of organs	protection, absorption
7	**(b)** cartilage	flexible
8	**(d)** smooth muscles	involuntary
9	**(a)** pivot joint	like a doorknob
10	**(d)** phloem cells	totipotency

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Chloroplasts (photosynthesis), chromoplasts (colour), leucoplasts (storage) (2).	2
12	Mitosis gives two identical cells for growth; meiosis gives four gametes with half the chromosomes (1+1).	1+1
13	A place where two bones meet (1); ball and socket, hinge (1).	1+1
14	Protects the heart and lungs and expands and contracts during breathing (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	All organisms are made of cells; the cell is the basic unit; cells come from cells (2); it unifies biology (1).	2+1
16	Free movement in many directions requires joints such as ball and socket (1.5); shoulder and hip, knee hinge (1.5).	1½+1½
17	Lipids made in SER, proteins on RER ribosomes (1.5); moved to the Golgi apparatus and carried by vesicles to the membrane (1.5).	1½+1½
	OR: In water the cells gain water and become turgid (1.5); in salt solution they lose water by osmosis and become flaccid (1.5).	1½+1½
18	Ribs are attached by flexible cartilage (1); they move to expand and contract the chest (1); so air moves in and out of the lungs (1).	1+1+1
	OR: Single phloem cells were grown in a medium with sugars and hormones (1.5); they dedifferentiated, formed a cell mass and grew into a whole plant, showing totipotency (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Biconcave, no nucleus, full of haemoglobin (2). (ii) Nerve cell, muscle cell or root hair cell with its function (2). (iii) About 120 days (1).	2+2+1
	OR: (i) 4 mm ÷ 20 = 0.2 mm = 200 µm (1). (ii) 100X (1). (iii) 200 µm × 100 = 20 000 µm = 20 mm (2). (iv) Resolution of the lenses (1).	1+1+2+1
20	(i) The ability of a cell to grow into a whole organism (1). (ii) Phloem cells dedifferentiated into a cell mass and developed into plantlets (3). (iii) Not easily; animal cells are far less totipotent (1).	1+3+1
	OR: (i) Collenchyma (1). (ii) Meristematic tissue (1). (iii) Xylem (1). (iv) Same size, same soil, same water and light (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.